

Shore Pods: A Small, Low-Cost Platform for Energy Harvesting and Maritime Sensing

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Abstract—Maritime sensing faces challenges from lack of energy and communications in remote regions. Especially when constrained by cost, sensors struggle to provide effective coverage of the maritime environment in ports and waterways. This work introduces a standardized, small, low-cost Shore Pod sensor platform for deployment in large numbers to autonomously monitor the maritime domain. Using this platform, we evaluate solar energy harvesting for moored Shore Pods. Given historical irradiance data of a target deployment environment and onboard sensor characteristics, we propose and evaluate an energy buffer sizing process to maximize uptime while minimizing volume, mass, and cost. We identify three energy-scheduling pitfalls and evaluate operating strategies to avoid these pitfalls and maximize uptime across operating environments. Top-performing Shore Pod operating strategies achieve 98.9% of maximum uptime.

Index Terms—shore pod, energy harvesting, maritime domain awareness, intermittent computing, sensor platform

I. INTRODUCTION

A Shore Pod is a small, standardized, low-cost sensor platform deployed in large numbers to autonomously monitor the maritime domain. The energy and communication constraints of remote regions limit maritime domain awareness (MDA). Shore Pods act as a system-of-systems for MDA, enabling ubiquitous remote sensing in, e.g., ports and waterways. The low-cost, distributed nature of Shore Pods makes individual devices attritable, which supports resilience in the sensing network. These sensor platforms process edge data autonomously to avoid communication bottlenecks, quickly notify operators of detected events via cellular or satellite links, and harvest energy for long-term operation. The standardized Shore Pod platform enables integration from multiple vendors, making sensor payloads interoperable and buoying the blue economy.

Shore Pods take inspiration from the CubeSat [1] standard in the space domain. Before CubeSats, satellites were massive, car-sized, and often cost over a billion dollars. CubeSats enable relatively low-cost proliferation of nanosatellites, develop a satellite engineering workforce in universities, and drive commercialization of space. In a similar way, Shore Pods aim to impact the maritime domain by increasing sensor coverage, educating ocean engineers, and broadening commercialization. While Argo floats (\$250k each) provide sparse sensing of the open ocean [2], [3], [4], [5], [6], moored Shore Pods provide dense sensing in ports and waterways at three orders of magnitude lower cost. This work answers the question:



Fig. 1. An implementation of a “1S” Shore Pod. Left: a 3D-printed hull. Right: electronics for energy harvesting and management, operations, and sensing.

How should an energy-harvesting maritime sensor schedule operations to maximize uptime when challenged by an energy deficit? To address this challenge, this work (i) proposes an energy buffer sizing process to maximize capacity while minimizing size and weight; (ii) identifies three energy-harvesting pitfalls: an energy buffer overflow, buffer depletion, and unnecessary hibernation; and (iii) evaluates four operating strategies — energy-blind, lazy, frugal, and generous — across five operating environments with unique energy characteristics.

In summary, this work contributes the following:

- An energy buffer sizing process to identify the smallest capacity Shore Pod supercapacitor or battery for a given target deployment environment (Section III).
- A quantitative characterization of three energy-harvesting pitfalls: energy buffer overflow, buffer depletion, and unnecessary hibernation (Section V-A).
- A comparison of four Shore Pod operating strategies — energy-blind, lazy, frugal, and generous — to address different energy-harvesting pitfalls (Section V-B).
- Evaluation of Shore Pod uptime in different energy-harvesting environments (e.g., energy-scarce vs. energy-abundant), showing that the generous operating strategy consistently outperforms other strategies (Section V-D).
- Results showing that these operating strategies achieve between 59% and 99% of the maximum uptime depending on the target operating environment.

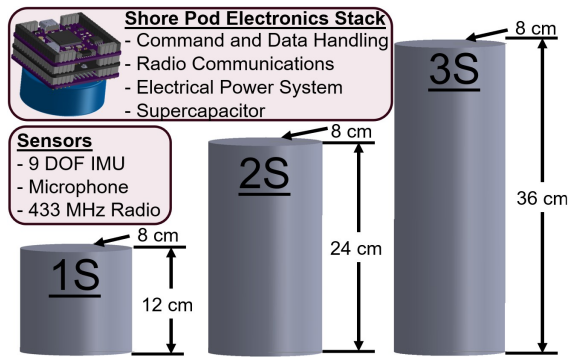


Fig. 2. Shore Pod form factors and electronics. A “1S” Shore Pod is a cylinder with an 8 cm diameter and a 12 cm height; 2S and 3S Shore Pods are taller.

II. BACKGROUND AND MOTIVATION

Shore Pods — small, standardized, low-cost, maritime sensor platforms — deploy in large numbers to autonomously monitor the maritime domain. The Shore Pod standard [7] defines multiple form factors; Figure 2 shows 1S, 2S, and 3S sizes. Each size shares an 8 cm diameter; a 1S Shore Pod measures 12 cm tall, a 2S is 24 cm, and a 3S is 36 cm. In the hull, a Shore Pod hosts electronics for the electrical power system, which manages energy harvesting, storage, and consumption, the radio for communication, and a command and data handling module to manage sensors. Unlike other maritime sensing platforms [8], [9], [10], [11], a Shore Pod harvests energy — a capability made possible by its low power draw and an intermittent-computing [12] operating strategy. This work considers solar; other energy-harvesting strategies may utilize piezoelectric fins or wave energy converters (WECs). Shore Pod electronics are open-source, modular, and customizable to support sensor payloads and energy-harvesting technologies for each mission. Figure 1 shows multiple sensor payloads: a micro-electromechanical system (MEMS) inertial measurement unit (IMU) to track orientation, a microphone for acoustic sensing, and a radio for receiving commands and transmitting data. This work focuses on moored Shore Pods for port and waterway MDA.

This low-size, weight, power, and cost (SWaP-C), commercial, off-the-shelf (COTS), distributed approach to sensing has seen success in domains outside the maritime environment. In space, the CubeSat standard [1] creates a market for COTS, space vehicle (SV) subsystems at lower SWaP-C than large, monolithic, one-of-a-kind satellites. The PocketQube standard [13] advances even smaller, lower-cost systems for space. These standards — focused on low-SWaP-C, COTS systems — boost the satellite engineering workforce, leading to Earth orbit proliferation and enabling trillion-dollar growth in space commercialization. Similarly, low-SWaP-C unmanned aerial vehicles (UAVs) (i.e., drones) change sensing in unstructured environments like construction, building inspection, disaster response, and search and rescue (SAR) [14], [15], [16], [17], [18]. Shore Pods advance low-SWaP-C, COTS, distributed, maritime sensing by training a digital-systems-savvy ocean engineering workforce to better bolster the blue economy.

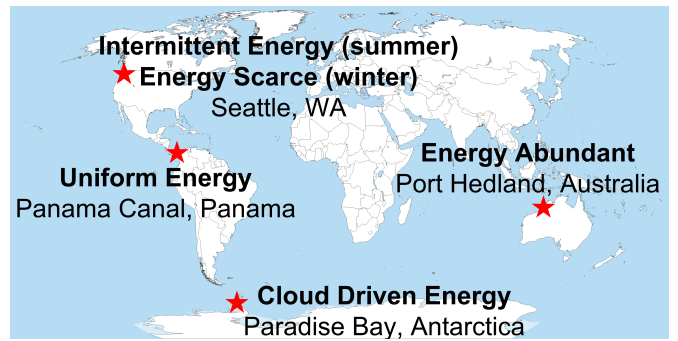


Fig. 3. Energy harvesting characteristics of each case-study location. Operation in Seattle during the summer gives a very different energy-harvesting profile than during the winter. Section IV provides further details.

Shore Pods join important work on small, cheap, highly capable, autonomous maritime sensing. The remote-controlled Glacial Frontal Ablation Monitoring Robot (GLAMOR) collects glacier terminus acoustic and visual data [8]. SeaPerch is a low-cost, educational autonomous underwater vehicle (AUV) for workforce development [9], [10]. The Acrobatic Low-cost Portable Hybrid AUV (ALPHA) [11] is a 30 kg, 1.5 m research platform for underwater robotics, similar to the Heterogeneous Swarm of Underwater Robotic Fish (H-SURF) [19]. While these ocean vehicles support some degree of autonomy, they all subscribe to a *mission-scale* concept-of-operations (CONOPS): the vehicle embarks on a sortie, performs tasks with stored energy (e.g., batteries), and returns to a controlling vehicle or location. Shore Pods are set-and-forget sensor platforms for widespread, long-term, autonomous deployment via energy harvesting with tolerance for attrition.

Motivating Case Study: The Argo Program

To motivate the Shore Pod approach to MDA, we compare with the long and storied Argo initiative. An Argo float is a free-floating buoy for ocean sensing; it drifts with currents and collects temperature and salinity data. Argo floats are large compared to Shore Pods: 12 cm in diameter and 1.2 m long. A float costs \$250k and has a single-use battery; Argo floats do not harvest energy. To operate for years, each float severely restricts data collection and transmission. An Argo float is large in size, weight, and cost while maintaining low power.

At end-of-life, Argo floats sink to the ocean floor. With about 4000 Argo floats operating globally, and given Earth’s $362 \times 10^6 \text{ km}^2$ of ocean, each float is responsible for $85\,000 \text{ km}^2$ — an area larger than Austria, which results in sparse coverage. The size and cost of an Argo float prevent port or waterway coverage — critical areas for MDA. Argo floats drift with currents, preventing targeted coverage.

Shore Pods cover a capability gap by providing dense, persistent sensor coverage of particular, targeted maritime areas with low-SWaP-C, high-count, attritable devices that harvest energy from the environment for long-term operation. Costing roughly three orders of magnitude less than an Argo float, a Shore Pod can be moored in ports and waterways to supply persistent maritime insights. Energy harvesting enables indefinite operations without retrieval or battery replacement.

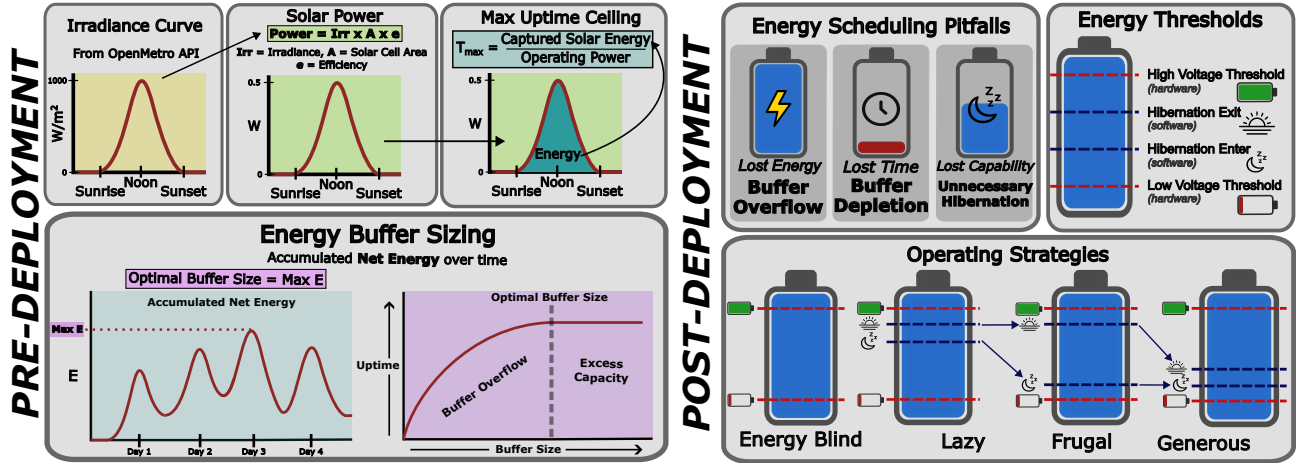


Fig. 4. Left: pre-deployment energy buffer sizing. Given expected irradiance and Shore Pod characteristics, determine maximum uptime. Integrating net power gives minimum buffer size to avoid energy buffer overflow. Right: post-deployment operating strategies avoid different energy-scheduling pitfalls.

III. THE SHORE POD ENERGY MANAGEMENT SYSTEM

A Shore Pod is a small, standardized, low-cost maritime sensor platform for deployment in large numbers to provide MDA. Low-SWaP-C makes energy harvesting feasible. However, the device often operates in an *energy deficit*: the rate of energy consumption exceeds the rate of energy harvesting. As a result, a Shore Pod transitions between active operation and charging. Factors such as the capacity of the onboard energy buffer (e.g., a battery or supercapacitor), charging entrance and exit thresholds, and hibernation decisions — coupled with energy harvesting environment characteristics — determine uptime. Poor choices reduce uptime, requiring deployment of more Shore Pods to achieve sensor coverage goals; intelligent energy management gives equal coverage with fewer devices. This work evaluates solar energy harvesting across five environments: cloud-driven, intermittent, abundant, scarce, and uniform energy; see Figure 3. At Shore Pod sizes, other energy-harvesting methods (e.g., WECs and piezoelectrics) provide more than an order of magnitude less power.

Shore Pod energy management has two phases: pre-deployment energy buffer sizing and post-deployment operation scheduling. Figure 4, left, shows pre-deployment sizing; Figure 4, right, shows post-deployment operation scheduling.

Pre-deployment: Energy Buffer Sizing

Prior to deployment, the energy management system selects an energy buffer (e.g., a battery or supercapacitor) capacity small enough to minimize SWaP-C and large enough to avoid operating inefficiencies such as an *energy buffer overflow*: when the system operates at an energy surplus (harvests energy faster than it consumes energy) but the buffer is full, creating an opportunity cost where harvested energy is lost. Figure 4, bottom left, shows that an energy buffer overflow occurs until a minimum capacity threshold, at which point uptime plateaus.

Given a location and period of interest, sizing establishes a maximum uptime $U_{\max} = \int_0^T P_{\text{solar}}(t) dt / \bar{P}_{\text{op}}$, where $P_{\text{solar}}(t)$ is power harvested at each time t during the period of interest, and $\bar{P}_{\text{op}}$ is average power draw during that period. Integrating harvested power during the period of interest gives total

energy harvested, and dividing by average power draw gives maximum uptime. Solar power $P_{\text{solar}}(t) = \eta A I_{\text{ir}}(t)$, where $I_{\text{ir}}(t)$ is irradiance (W/m^2), A is active panel area (m^2), and η is efficiency. The Open-Meteo API [20] gives irradiance (see Section IV). Figure 4, top left, shows this procedure.

Given maximum uptime $U_{\max}$, the sizing procedure selects the smallest energy buffer capacity that avoids energy buffer overflow. By integrating *net* power (harvested power minus power consumed) with a floor at zero (a buffer cannot store negative energy), peak accumulated energy sets the minimum capacity to avoid overflow. See Figure 4, bottom left.

Post-deployment: Energy-aware Operating Strategies

We identify energy-scheduling pitfalls that lower Shore Pod uptime U below maximum uptime $U_{\max}$ and establish candidate operating strategies to avoid these pitfalls; Section V evaluates each of these operating strategies.

Energy buffer overflow occurs when the energy buffer fills to its capacity, causing subsequent surplus energy to be discarded and permanently reducing uptime by the amount of energy lost to overflow. *Buffer depletion* occurs when the energy buffer discharges below the hardware low-voltage threshold, forcing a pause in operations until the buffer charges to the hardware high-voltage threshold and causing downtime that cannot be recovered. *Unnecessary hibernation* occurs when the Shore Pod pauses operations despite sufficient energy to continue, wasting uptime. Figure 4, top right, shows each pitfall.

We evaluate how several operating strategies — energy-blind, lazy, frugal, and generous — avoid (or fail to avoid) each pitfall. Energy-blind, the state-of-the-art (SotA) (e.g., [21], [22], [12]) method, uses a fixed duty cycle without buffer state awareness; hardware low- and high-voltage thresholds force long recharge periods. In addition to hardware low- and high-voltage thresholds, the lazy, frugal, and generous strategies introduce hibernation entrance and exit thresholds. Figure 4, top right, shows these hardware and hibernation thresholds. Hibernation thresholds operate similarly to hardware thresholds — the Shore Pod sleeps at the entrance threshold and wakes at the exit threshold — but these thresholds can be set

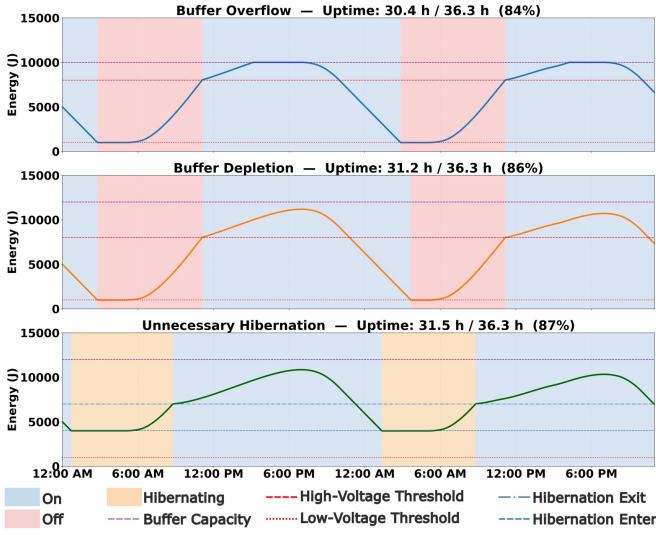


Fig. 5. Top: When the buffer reaches full capacity, surplus energy cannot be stored, reducing uptime. Middle: Reaching the low-voltage threshold requires charging until the high-voltage threshold, resulting in extended downtime. Bottom: Hibernating with energy available unnecessarily forgoes uptime.

to any value above the hardware low-voltage threshold. The lazy operating strategy sets both the hibernation entrance and exit thresholds high in the buffer to avoid buffer depletion. The frugal operating strategy sets the hibernation entrance threshold low in the buffer to use more energy while keeping the hibernation exit threshold high to allow full charges. The generous operating strategy sets both the hibernation entrance and exit thresholds low in the buffer to use more energy and charge quickly. Figure 4, bottom right, shows each strategy.

IV. IMPLEMENTATION AND METHODOLOGY

We evaluate operating strategies in simulation; see Figure 1 for validation hardware. Our software artifact is available open-source at github.com/starbelt/shore-pod-OCEANS26. We evaluate five locations in 2025: Seattle, June 9-22 (intermittent energy) and December 1-14 (energy scarce); Panama Canal, March 1-14 (uniform energy); Paradise Bay, Antarctica, Dec 1-14 (cloud-driven energy); and Port Hedland, Australia, Jan 22-Feb 4 (energy abundant).

At each location, the Shore Pod begins with an energy buffer capacity set by pre-deployment buffer sizing. The simulation runs through a two-week solar power trace with 1 min time steps and updates buffer energy level and Shore Pod operating state given net power, buffer capacity, and strategy thresholds.

The simulator gives uptime for each operating strategy across all locations. Energy-blind uses a 50% duty cycle (six hours on, six hours off) with no buffer state awareness and subject to hardware low- and high-voltage thresholds; thus, the Shore Pod rarely operates for six hours continuously. During hardware-enforced charge periods, the device is off and cannot operate until reaching the high-voltage threshold. Lazy, frugal, and generous operate continuously and use hibernation thresholds to manage charge: the Shore Pod enters hibernation when the buffer falls to the entrance threshold and resumes when it recovers to the exit threshold. Lazy sets high entrance

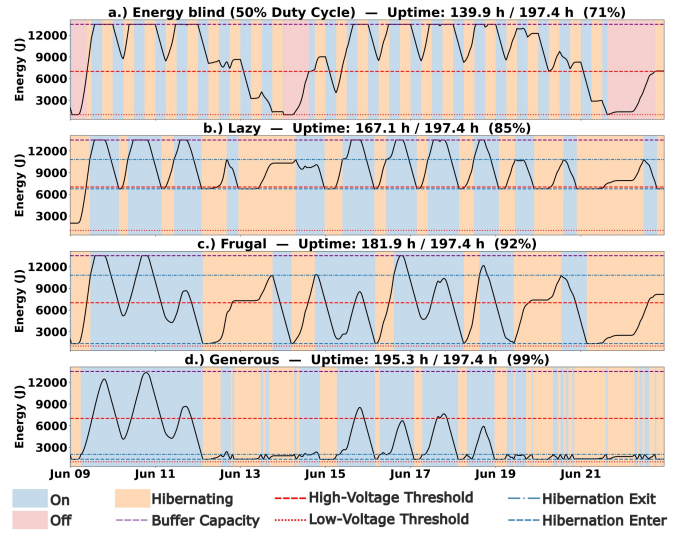


Fig. 6. Energy-blind (a) experiences energy buffer overflow, buffer depletion, and unnecessary hibernation. Lazy (b) avoids buffer depletion but experiences buffer overflow and unnecessary hibernation. Frugal (c) reduces overflows and unnecessary hibernation. Generous (d) achieves the most uptime.

and exit thresholds (50% and 80% of capacity, respectively). Frugal lowers the entrance threshold to 10% of capacity while keeping the exit threshold high at 80%. Generous lowers both thresholds, setting entrance at 10% and exit at 15% of capacity.

To evaluate the entire threshold space, we sweep 5050 hibernation entrance and exit combinations (see Figure 7); values sweep from 5% to 95% of the buffer size with a minimum gap of 50 J between entrance and exit thresholds.

V. EVALUATION AND RESULTS

Subsection V-A quantifies the impact of each energy-scheduling pitfall: energy buffer overflow, buffer depletion, and unnecessary hibernation. Subsection V-B evaluates the energy-blind, lazy, frugal, and generous operating strategies. These strategies are instances of an exhaustive search across hibernation entrance and exit thresholds (Subsection V-C). Subsection V-D records total uptime across locations and times (cloud-driven, intermittent, scarce, abundant, and uniform energy availability) compared to the maximum uptime.

A. Energy-scheduling Pitfalls Reduce Uptime

Figure 5 shows each energy-harvesting pitfall: energy buffer overflow, buffer depletion, and unnecessary hibernation. For clarity, we limit the period of interest to a subset of the intermittent energy case study: Seattle during a two-day (June 9-10) summer period (as opposed to the full, two-week case study). Figure 5 shows energy stored at each moment during the period of interest. Hardware low- and high-voltage thresholds (set by, e.g., the buck-boost converter and voltage detector) are shown with red, dashed lines. Hibernation entrance and exit thresholds (set by system software) are shown with blue, dashed lines. After reaching the hardware low-voltage threshold, the Shore Pod cannot operate until charged to the high-voltage threshold. After entering hibernation, the Shore Pod resumes operation once charged to the exit threshold.

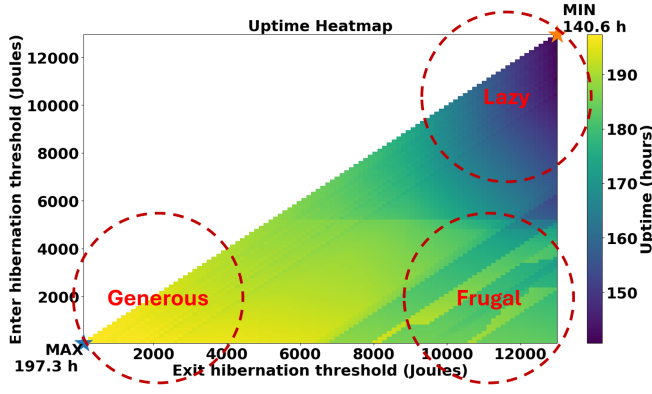


Fig. 7. Shore Pod uptime for 5050 operating strategies as a function of hibernation entrance (y-axis) and exit (x-axis) thresholds. Top-right: lazy strategies; bottom-right: frugal strategies; bottom-left: generous strategies.

Energy buffer overflow. Figure 5, top, measures uptime lost when the buffer fills during power surplus. When the Shore Pod buffer is at capacity while harvested energy rate exceeds expended energy rate, excess energy is lost. Because a Shore Pod operates at an energy deficit on a long enough time scale, energy lost to a buffer overflow lowers overall uptime. Figure 5, top, shows buffer overflows at plateaus where stored energy reaches buffer capacity (here, 10 kJ). Later, when the Shore Pod operates in an energy deficit, the buffer reaches the low-voltage threshold sooner due to lost energy. The Shore Pod achieves an uptime of 30.4 h, 84% of the maximum.

Energy buffer depletion. Figure 5, middle, measures uptime lost to buffer depletion. Raising buffer capacity (here, to 12 kJ) avoids energy buffer overflow. At the hardware low-voltage threshold, the Shore Pod must charge to the hardware high-voltage threshold; thresholds are set by, e.g., the buck-boost converter and voltage detector. Downtime cannot be recovered, and inactivity due to charging hurts uptime. Figure 5, middle, shows charge periods in red taking about 8 h. The Shore Pod achieves a total uptime of 31.2 h, 86% of the maximum.

Unnecessary hibernation. Figure 5, bottom, measures uptime lost to unnecessary hibernation. Hibernating before reaching the hardware low-voltage threshold avoids buffer depletion. However, “sleeping” too soon or waiting too long to resume operations wastes uptime when sufficient charge exists. Figure 5, bottom, shows hibernation periods in orange taking about 6 h despite energy availability. The Shore Pod achieves a total uptime of 31.5 h, 87% of the maximum.

B. Operating Strategies Avoid Energy-scheduling Pitfalls

Figure 6 measures energy-blind, lazy, frugal, and generous performance. We provide detailed plots for the intermittent energy case study; Figure 8 summarizes results for all locations and times. Figure 6 shows energy stored in the Shore Pod buffer at each time during the period of interest. Red lines label hardware low- and high-voltage thresholds, and blue lines label hibernation entrance and exit thresholds. A blue background indicates the Shore Pod is on, a red background that it is off, and an orange background that it is hibernating. Each plot marks buffer capacity (e.g., 12 kJ) set by buffer sizing.

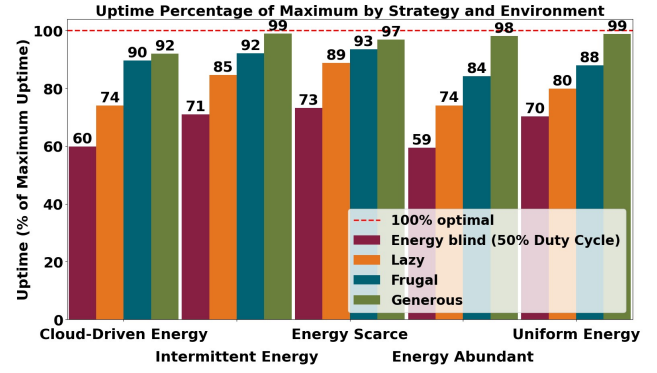


Fig. 8. The percentage of maximum uptime achieved by each operating strategy across five deployment environments. The generous strategy consistently outperforms energy-blind, lazy, and frugal strategies.

Energy-blind. This strategy operates the Shore Pod at a fixed, 50% duty cycle with no buffer state awareness. As a result, it experiences each energy-scheduling pitfall: energy buffer overflow, buffer depletion, and unnecessary hibernation. Figure 6 (a) shows buffer overflows when energy plateaus at capacity. Due to energy lost during overflow, the Shore Pod experiences buffer depletions while operating at energy deficits. Depletions trigger an “off” period while the Shore Pod charges. A fixed duty cycle forces the Shore Pod to hibernate despite sufficient energy to operate, further reducing uptime. For the intermittent energy case during Seattle’s summer, an energy-blind Shore Pod operates for 140 h at 71% of maximum uptime. Figure 8 gives results for all case studies.

Lazy. This strategy introduces hibernation thresholds to avoid buffer depletion but, due to unnecessary hibernation, still experiences buffer overflows. Figure 6 (b) shows fewer buffer overflows and no buffer depletions. This particular implementation of Lazy sets hibernation entrance and exit thresholds to 50% and 80% of buffer capacity, respectively (both are relatively high in the buffer); Figure 7 exhaustively sweeps all thresholds. A high entrance threshold guards against buffer depletion, but a high exit threshold fails to sufficiently guard against buffer overflow. A high entrance threshold also creates unnecessary hibernation while the buffer has energy to support operation. For the intermittent energy case during Seattle’s summer, a lazy Shore Pod operates for 167 h at 85% of maximum uptime. Figure 8 gives results for all case studies.

Frugal. This strategy lowers hibernation entrance thresholds to avoid unnecessary hibernation, but a high exit threshold creates energy buffer overflows. Figure 6 (c) shows fewer hibernation periods. This implementation of Frugal sets hibernation entrance and exit thresholds to 10% and 80% of buffer capacity, respectively; Figure 7 sweeps all thresholds. A low entrance threshold guards against unnecessary hibernation, but a high exit threshold fails to prevent buffer overflow. In the intermittent energy case, a frugal Shore Pod operates for 182 h (92% of maximum uptime). Figure 8 gives results for all cases.

Generous. This strategy lowers hibernation exit thresholds to avoid energy buffer overflows experienced by other strategies. Figure 6 (d) shows fewer overflows. Hibernation count

increases, but total downtime decreases. This implementation of Generous sets hibernation entrance and exit thresholds to 10% and 15% of buffer capacity, respectively (both are low in the buffer); Figure 7 sweeps all thresholds. A low entrance threshold prevents unnecessary hibernation, and a low exit threshold prevents buffer overflow. In the intermittent energy case, a generous Shore Pod operates for 195 h at 99% of maximum uptime. Figure 8 gives results for all case studies. Importantly, no operating strategy can obtain 100% of maximum uptime because the hardware low-voltage threshold effectively “locks” some energy from use.

C. Exhaustive Exploration of Operating Strategies

Subsection V-B gives results for specific realizations of each strategy in the intermittent energy case. Results are sensitive to hibernation entrance and exit thresholds. This subsection considers a full sweep of hibernation entrance and exit thresholds applied to the intermittent case study; results for all scenarios (cloud-driven, scarce, abundant, and uniform energy availability) are summarized in Subsection V-D. This sweep evaluates 5050 operating strategies with results in Figure 7.

The y -axis of Figure 7 sweeps hibernation entrance thresholds, and the x -axis sweeps hibernation exit thresholds; both range from 5% to 95% of buffer capacity with at least 50 J between entrance and exit thresholds. Colors indicate uptime: yellow is higher (better), purple is lower (worse), with green and blue for intermediate values. Strategies at the top-right are Lazy; the bottom-right is Frugal; the bottom-left is Generous.

The middle of Figure 7 shows a distinct, horizontal line at an entrance threshold of about 5.5 kJ, indicating an uptime drop for all exit thresholds. When hibernation entrance reaches this value or higher, the energy buffer no longer contains sufficient capacity for the Shore Pod to operate through the night. Setting a hibernation entrance threshold lower gives sufficient energy for the Shore Pod to operate through the night without hibernation, increasing uptime. Diagonal patterns at the lower-right correspond to the number of charging cycles per period. Bands of lower uptime occur when entrance and exit thresholds result in charging from hibernation entrance to hibernation exit that cannot be fulfilled by energy from N days and instead requires $N + 1$ days of energy harvesting. Minimum uptime occurs in the top-right (Lazy, 141 h), and maximum uptime occurs in the bottom-left (Generous, 197 h).

D. Generous Maximizes Uptime across Deployments

We measure uptime for each operating strategy (energy-blind, lazy, frugal, and generous) across cloud-driven, intermittent, scarce, abundant, and uniform energy case studies. Figure 3 shows each location selected for distinct irradiance traces. Figure 8 quantifies the percentage of maximum uptime achieved by each strategy across all deployments.

The generous operating strategy consistently provides the most uptime, ranging from 92% to 99% of maximum uptime. Frugal achieves the second-highest uptime across deployment environments, ranging from 84% to 93% of maximum uptime. Lazy achieves the third-highest uptime across deployment

environments, ranging from 74% to 89% of maximum uptime. Energy-blind achieves the lowest uptime across deployment environments, ranging from 59% to 73% of maximum uptime.

Generous gives its best performance in intermittent and uniform energy environments; the lowest uptime fraction occurs in a cloud-driven energy environment. Frugal gives its best performance in an energy-scarce environment (with generous providing a higher uptime); the lowest uptime fraction occurs in an energy-abundant environment. Lazy gives its best performance in an energy-scarce environment (with generous providing a higher uptime); the lowest uptime fraction occurs in cloud-driven and abundant energy environments. Energy-blind gives its best performance in an energy-scarce environment (with generous providing a higher uptime); the lowest uptime fraction occurs in an energy-abundant environment.

VI. RELATED WORK

Recent work addresses maritime sensing, low-cost platforms, and energy harvesting separately. Hybrid robotic networks combine AUVs, surface vehicles, and fixed sensors [23], [24]; deep learning detects and classifies targets from sonar [25], [26] and drone imagery [27]; AUVs inspect infrastructure [28]; and AUV-aided clustering reduces sensor-network transmission energy [29], [30]. Low-cost, educational platforms such as SeaPerch [9], [10], ALPHA [11], and compact surface vehicles [8] prioritize affordability and remote-control simplicity over unattended, long-duration autonomy. Maritime energy harvesting spans WECs [31], breakwater-integrated [32] and triboelectric [33] harvesters, floating and port-scale photovoltaics [34], [35], hybrid-power surface vehicles [36], and UAV-maintained offshore arrays [37]. For intermittent computing, reconfigurable energy storage [12], power-aware heterogeneous architectures [38], year-long under-ice AUV deployments [39], and cooperative low-cost robot swarms [19] address energy-aware operation under scarcity. None of these approaches alone provides dense, persistent, unattended MDA; Shore Pods close this gap with a standardized, low-SWaP-C, attritable platform combining solar harvesting with the energy buffer sizing and operating strategies.

VII. CONCLUSION

This work introduces Shore Pods: a standardized, small, low-cost sensor platform for dense, persistent, and autonomous maritime sensing. Shore Pods deploy widely and cost three orders of magnitude less than Argo floats. Low-SWaP-C makes energy harvesting practical but requires careful energy buffer sizing and intelligent operating strategies. We quantify the impact of energy-harvesting pitfalls across deployments. We evaluate lazy, frugal, and generous operating strategies against an energy-blind approach. The generous operating strategy consistently provides the highest fraction (between 92% and 99%) of maximum uptime.

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REFERENCES

- [1] T. C. Program, "Cubesat design specification rev. 14.1," tech. rep., California Polytechnic State University, San Luis Obispo, 2022.
- [2] V. Thierry, H. Claustre, O. Pasquero de Fommervault, N. Zilberman, K. S. Johnson, B. A. King, S. E. Wijffels, U. T. Bhaskar, M. A. Balmaseda, M. Belbeoch, *et al.*, "Advancing ocean monitoring and knowledge for societal benefit: the urgency to expand argo to oneargo by 2030," *Frontiers in Marine Science*, 2025.
- [3] T. Morris, M. Scanderbeg, D. West-Mack, C. Gourcuff, N. Poffa, T. U. Bhaskar, C. Hanstein, S. Diggs, L. Talley, V. Turpin, *et al.*, "Best practices for core argo floats-part 1: getting started and data considerations," *Frontiers in Marine Science*, 2024.
- [4] T. Morris, M. Scanderbeg, D. West-Mack, C. Gourcuff, N. Poffa, T. V. Udaya Bhaskar, C. Hanstein, S. Diggs, L. Talley, V. Turpin, *et al.*, "Best practices for core argo floats-part 2: physical handling, deployment and metadata considerations," *Frontiers in Marine Science*, 2024.
- [5] G. C. Johnson and A. J. Fassbender, "After two decades, argo at pmel, looks to the future," *Oceanography*, 2023.
- [6] D. Roemmich, G. C. Johnson, S. Riser, R. Davis, J. Gilson, W. B. Owens, S. L. Garzoli, C. Schmid, and M. Ignaszewski, "The argo program: Observing the global ocean with profiling floats," *Oceanography*, 2009.
- [7] A. Skotak and B. Denby, "Shore pod standard," tech. rep., Virginia Tech, 2025.
- [8] H. Vishnu, M. Chitre, B. Kalyan, T. S. Pieng, D. Stokes, E. Weidner, M. Hoffmann-Kuhnt, and O. Glowacki, "A low-cost surface vehicle for studying the near-terminus region at tidewater glaciers," in *IEEE OCEANS*, 2025.
- [9] A. Garcia-Langle, I. Alvarez, M. Sarno, D. Brancazio, A. Bennett, and M. S. Triantafyllou, "Sea perch ii: Educational autonomous underwater vehicle for k-12+ audiences," in *IEEE OCEANS*, 2023.
- [10] A. Garcia-Langle, I. Alvarez, A. Chen, A. Li, H. Wang, D. Brancazio, V. Gutierrez, A. Bennett, and M. Triantafyllou, "Development of educational marine soft robotics stem platform as new iteration of seaperch k-12 national outreach program," in *IEEE OCEANS*, 2022.
- [11] M. Zhou, E. C. Gezer, W. McConnell, and C. Yuan, "Acrobatic low-cost portable hybrid auv (alpha): System design and preliminary results," in *IEEE OCEANS*, 2022.
- [12] A. Colin, E. Ruppel, and B. Lucia, "A reconfigurable energy storage architecture for energy-harvesting devices," in *ACM Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2018.
- [13] S. Radu, M. Uludag, S. Speretta, J. Bouwmeester, A. Dunn, T. Walkinshaw, P. Kaled Da Cas, and C. Cappelletti, "The pocketcube standard issue 1," tech. rep., TU Delft, 2018.
- [14] E. Sie, Z. Liu, and D. Vasisht, "Batmobility: Towards flying without seeing for autonomous drones," in *ACM International Conference on Mobile Computing and Networking (MobiCom)*, 2023.
- [15] M. Jeong, M. Roznere, S. Lensgraf, A. Sniffen, D. Balkcom, and A. Q. Li, "Catabot: Autonomous surface vehicle with an optimized design for environmental monitoring," in *IEEE OCEANS*, 2020.
- [16] J. Irizarry and D. B. Costa, "Exploratory study of potential applications of unmanned aerial systems for construction management tasks," *Journal of Management in Engineering*, 2016.
- [17] C. A. F. Ezequiel, M. Cua, N. C. Libatique, G. L. Tagonan, R. Alampay, R. T. Labuguen, C. M. Favila, J. L. E. Honrado, V. Canos, C. Devaney, *et al.*, "Uav aerial imaging applications for post-disaster assessment, environmental management and infrastructure development," in *IEEE International Conference on Unmanned Aircraft Systems (ICUAS)*, 2014.
- [18] C. Eschmann, "Unmanned aircraft systems for remote building inspection and monitoring," 2012.
- [19] S. Iacoponi, G. J. Van Vuuren, G. Santaera, N. Mankovskii, I. Zhilin, F. Renda, C. Stefanini, and G. De Masi, "H-surf: Heterogeneous swarm of underwater robotic fish," in *IEEE OCEANS*, 2022.
- [20] Open-Meteo, "Historical weather api documentation," <https://open-meteo.com/en/docs/historical-weather-api>, 2024.
- [21] K. Maeng and B. Lucia, "Adaptive low-overhead scheduling for periodic and reactive intermittent execution," in *ACM Conference on Programming Language Design and Implementation (PLDI)*, 2020.
- [22] B. Denby and B. Lucia, "Orbital edge computing: Nanosatellite constellations as a new class of computer system," *ACM Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2020.
- [23] G. Ferri, A. Faggiani, P. Stinco, A. Tesei, and R. Been, "Cooperation in surveillance autonomous underwater networks: field experience from repmus22," in *IEEE OCEANS*, 2023.
- [24] G. Ferri, R. Grasso, A. Faggiani, F. de Rosa, E. Camossi, A. Grati, P. Stinco, A. Tesei, R. Been, K. D. LePage, *et al.*, "A hybrid robotic network for maritime situational awareness: Results from the infore22 sea trial," in *IEEE OCEANS*, 2022.
- [25] L. Annamalai, C. Benjamin, *et al.*, "Pingcnn: A low-latent and low-cost network for mine detection in side scan sonar signal," *IEEE Journal of Oceanic Engineering (JOE)*, 2026.
- [26] Y. Song, B. He, and P. Liu, "Real-time object detection for auvs using self-cascaded convolutional neural networks," *IEEE Journal of Oceanic Engineering (JOE)*, 2019.
- [27] F. Zhao, D. Xi, Y. Chen, B. Ma, Y. Liu, J. Wang, and K. Mizuno, "Basic study of deep learning based efficient hermit crabs detection from drone-captured images," in *IEEE OCEANS*, 2024.
- [28] S. Chun, H. Yokohata, M. Ohashi, K. Ohkuma, S. Ito, S. Hirabayashi, and T. Maki, "3d detection and tracking of mooring lines of floating offshore wind turbines by autonomous underwater vehicle," in *IEEE OCEANS*, 2024.
- [29] Z. Hao, W. Li, and Q. Zhang, "Efficient clustering data collection in auv-aided underwater sensor network," in *IEEE OCEANS*, 2023.
- [30] R. Zhao, C. Liu, and X. Shen, "Energy-balanced clustering protocols for underwater sensor networks," in *IEEE OCEANS*, 2024.
- [31] R. G. Coe, G. Bacelli, D. Gaebele, A. Keow, and D. Forbush, "Co-design of a wave energy converter through bi-conjugate impedance matching," *Mechatronics*, 2025.
- [32] A. N. Mullen, P. M. Koola, and S. S. Girimaji, "Wave energy converter integrated with breakwater: Experimental investigation of hydrodynamic efficiency," in *IEEE OCEANS*, 2023.
- [33] A. M. Tshimbu, T. Zou, and L. Zhang, "Advancements and challenges in triboelectric nanogenerator technology for ocean wave energy harvesting," *IEEE OCEANS*, 2024.
- [34] R. Kanotra and R. Shankar, "Floating solar photovoltaic mooring system design and analysis," in *IEEE OCEANS*, 2022.
- [35] S. Bali, S. Thangalakshmi, and R. Balaji, "Renewable energy options for seaports," in *IEEE OCEANS*, 2022.
- [36] J. Wang, J. Wang, Z.-Y. Liu, X.-F. Liang, and H. Yi, "Design and experiments of a sustainable hybrid power unmanned surface vehicle for maritime patrol," in *IEEE OCEANS*, 2024.
- [37] R. Morais, J. J. Martins, P. Lima, A. Dias, A. Martins, J. Almeida, and E. Silva, "Novel approach for offshore photovoltaic panels inspection with vtol uav," in *IEEE OCEANS*, 2024.
- [38] H. Desai and B. Lucia, "A power-aware heterogeneous architecture scaling model for energy-harvesting computers," *IEEE Computer Architecture Letters (CAL)*, 2020.
- [39] A. B. Phillips, G. Salavasidis, R. Templeton, C. Duguid, R. Marlow, A. Martin, M. Kingsland, R. Sherif, F. Fanelli, D. Fenucci, *et al.*, "Anchoring auvs: towards a year-long deployment under sea ice," in *IEEE OCEANS*, 2025.